CO2 ASSESSMENT TOOL 2 Scenrio based questions

SIMATS ENGINEERING

SIMATS Online Programming Lab

C PROGRAMMING

RESHINI PRIYA B
192320035RB

CO2 - AT2 - SCENARIO BASED QUESTIONS

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QUESTIONS

Q1
Scenario Based - Smart Waste Recycling Plant System
25 marks

Q2
Scenario Based - IT Company Project Performance Evaluation (2025–2026).
25 marks

Q3
Scenario Based - 3. Semiconductor Chip Manufacturing System
25 marks

Q4
Scenario Based - 4. Cloud Storage Subscription Management (2026)
25 marks

4/4 answered

Q1 **Scenario Based - Smart Waste Recycling Plant System** 25 marks

A recycling plant processes 5,000 kg of waste every day.
Waste Categories:
Plastic
Paper
Metal
Electronic Waste
Efficiency Levels:
≥90% Recycled → Excellent
75–89% → Good
50–74% → Average
Below 50% → Poor
(a) Develop a C program using arithmetic operators to calculate recycled waste percentage and remaining waste. Use decision-making statements to determine plant efficiency.
(b) Use looping constructs to process waste data for 30 days. Use continue to skip maintenance days and break if the processing machine fails.

Your Code

```c
#include <stdio.h>

int main() {

    float totalWaste = 5000.0;
    float recycledWaste, remainingWaste, efficiencyPercent;

    printf("Enter the amount of recycled waste (in kg) out of %.2f kg:
    scanf("%f", &recycledWaste);


    remainingWaste = totalWaste - recycledWaste;
    efficiencyPercent = (recycledWaste / totalWaste) * 100;

    printf("\n--- Waste Processing Report ---\n");
    printf("Total Waste: %.2f kg\n", totalWaste);
    printf("Recycled Waste: %.2f kg\n", recycledWaste);
    printf("Remaining Waste: %.2f kg\n", remainingWaste);
    printf("Recycled Percentage: %.2f%%\n", efficiencyPercent);


    if (efficiencyPercent >= 90) {
        printf("Plant Efficiency: Excellent\n");
    } else if (efficiencyPercent >= 75) {
        printf("Plant Efficiency: Good\n");
    } else if (efficiencyPercent >= 50) {
        printf("Plant Efficiency: Average\n");
    } else {
        printf("Plant Efficiency: Poor\n");
    }


    printf("\n--- 30 Day Waste Processing Log ---\n");
    int day;
    int machineFailed = 0;

    for (day = 1; day <= 30; day++) {

        if (day % 10 == 0) {
            printf("Day %d: Maintenance Day - Skipping processing.\n",
            continue;
        }


        int status;
        printf("Day %d: Enter machine status (1 = Working, 0 = Failed)
        scanf("%d", &status);

        if (status == 0) {
            printf("Day %d: Machine failed! Stopping waste processing.
            machineFailed = 1;
            break;
        }

        printf("Day %d: Waste processed successfully.\n", day);
    }

    if (!machineFailed) {
        printf("\nAll 30 days completed without machine failure.\n");
    } else {
        printf("\nProcessing stopped early due to machine failure.\n");
    }

    return 0;
}
```

Input

```
4500
1 1 1 1 1 1 1 1 1
1 1 1 1 1 1 1 1 1
1 1 1 1 1 1 1 1 1
```

Output

Run Save

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Q2 Scenario Based - IT Company Project Performance Evaluation (2025–2026).

25 marks

An IT company completed 20 software projects during the financial year 2025–2026. Each project is evaluated based on Quality Score (100 marks) and Delivery Delay (days).

Performance Criteria:

Quality ≥ 90 and Delay ≤ 2 days → Excellent

Quality 75–89 and Delay ≤ 5 days → Good

Quality 60–74 → Average

Below 60 → Needs Improvement

A) Develop a C program using arithmetic operators to calculate the average quality score and delivery percentage. Use decision-making statements to classify each project.

B) Use looping constructs to process all 20 projects. Use continue to ignore cancelled projects and break if management stops the evaluation.

Your Code

```c
#include <stdio.h>

int main() {
 int n = 20;
 float quality[20], delay[20];
 float totalQuality = 0, totalDelay = 0;
 int cancelled[20]; // 1 = cancelled, 0 = not cancelled
 int stopEvaluation;
 int count = 0; // number of projects actually evaluated

 printf("Enter details for %d projects:\n", n);

 for (int i = 0; i < n; i++) {
 printf("\nProject %d\n", i + 1);

 printf("Is this project cancelled? (1 = Yes, 0 = No): ");
 scanf("%d", &cancelled[i]);
 if (cancelled[i] == 1) {
 printf("Project %d is cancelled. Skipping...\n", i + 1);
 continue;
 }

 printf("Does management want to stop evaluation? (1 = Yes, 0 = No): "
 scanf("%d", &stopEvaluation);

 if (stopEvaluation == 1) {
 printf("Evaluation stopped by management.\n");
 break;
 }
 printf("Enter Quality Score (0-100): ");
 scanf("%f", &quality[i]);

 printf("Enter Delivery Delay (days): ");
 scanf("%f", &delay[i]);

 // Accumulate totals using arithmetic operators
 totalQuality = totalQuality + quality[i];
 totalDelay = totalDelay + delay[i];
 count++;
 printf("Classification: ");
 if (quality[i] >= 90 && delay[i] <= 2) {
 printf("Excellent\n");
 }
 else if (quality[i] >= 75 && quality[i] <= 89 && delay[i] <= 5) {
 printf("Good\n");
 }
 else if (quality[i] >= 60 && quality[i] <= 74) {
 printf("Average\n");
 }
 else {
 printf("Needs Improvement\n");
 }
 }

 // Calculate averages using arithmetic operators
 if (count > 0) {
 float avgQuality = totalQuality / count;
 float avgDelay = totalDelay / count;
 float deliveryPercentage = (avgDelay <= 5) ? 100.0 - (avgDelay * 2) :
 printf("\n---- Summary ----\n");
 printf("Total Projects Evaluated: %d\n", count);
 printf("Average Quality Score: %.2f\n", avgQuality);
 printf("Average Delivery Delay: %.2f days\n", avgDelay);
 printf("Estimated Delivery Percentage: %.2f%%\n", deliveryPercentage)
 } else {
 printf("\nNo projects were evaluated.\n");
 }

 return 0;
}
```

Input

```
0 0 95 1.5
0 0 82 3.0
1
0 0 55 6.0
```

Output

Run Save

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- Q2 Scenario Based - IT Company Project Performance Evaluation (2025–2026). — 25 marks
- Q3 Scenario Based - 3. Semiconductor Chip Manufacturing System — 25 marks
- Q4 Scenario Based - 4. Cloud Storage Subscription Management (2026) — 25 marks

4/4 answered

Q3 Scenario Based - 3. Semiconductor Chip Manufacturing System

25 marks

A semiconductor company manufactures 10,000 chips per day.

Quality Standards:

Defect Rate <1% → Grade A

1–3% → Grade B

3–5% → Grade C

Above 5% → Reject Batch

A) Develop a C program using arithmetic operators to calculate the defect percentage and accepted chips. Use decision-making statements to assign quality grades.

B) Use looping constructs to process production for 30 days. Use continue for maintenance days and break if a machine breakdown occurs.

Your Code

```c
#include <stdio.h>

int main() {
    int totalChips = 10000;
    int defectiveChips;
    float defectPercentage;
    int acceptedChips;
    int day;
    int maintenanceDay = 10;
    int breakdownDay = 25;

    printf("=== Semiconductor Chip Manufacturing - 30 Day Production Re

    for (day = 1; day <= 30; day++) {


        if (day == maintenanceDay) {
            printf("Day %d: Maintenance Day - Production Skipped.\n",
            continue;
        }


        if (day == breakdownDay) {
            printf("Day %d: Machine Breakdown Occurred! Stopping Produ
            break;
        }


        defectiveChips = (day * 37) % 600;


        defectPercentage = ((float)defectiveChips / totalChips) * 100;
        acceptedChips = totalChips - defectiveChips;

        printf("\nDay %d:\n", day);
        printf("Total Chips Manufactured : %d\n", totalChips);
        printf("Defective Chips          : %d\n", defectiveChips);
        printf("Defect Percentage        : %.2f%%\n", defectPercentage
        printf("Accepted Chips           : %d\n", acceptedChips);


        if (defectPercentage < 1) {
            printf("Quality Grade            : A\n");
        } else if (defectPercentage >= 1 && defectPercentage <= 3) {
            printf("Quality Grade            : B\n");
        } else if (defectPercentage > 3 && defectPercentage <= 5) {
            printf("Quality Grade            : C\n");
        } else {
            printf("Quality Grade            : REJECTED BATCH\n");
        }
    }

    printf("\n=== Production Monitoring Ended ===\n");

    return 0;
}
```

Input

stdin input

Output

Run Save

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25 marks

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25 marks

Q4 Scenario Based - 4. Cloud Storage Subscription Management (2026)
25 marks

4/4 answered

Q4 Scenario Based - 4. Cloud Storage Subscription Management (2026)

25 marks

A cloud company offers storage plans:

Basic – 100 GB

Standard – 500 GB

Enterprise – 2 TB

Extra storage costs ₹4 per GB.

If storage usage exceeds 95%, a warning should be generated.

A) Develop a C program using arithmetic operators to calculate used storage, remaining storage, and extra charges. Use decision-making statements to determine storage status.

B) Use looping constructs to process multiple customers. Use continue for inactive accounts and break if server storage becomes full.

Your Code

```c
#include <stdio.h>
#include <string.h>

int main() {
    int n;
    printf("Enter number of customers: ");
    scanf("%d", &n);

    float serverCapacity = 5000.0;
    float serverUsed = 0.0;

    for (int i = 0; i < n; i++) {
        char plan[20];
        float usedGB;
        int isActive;

        printf("\nCustomer %d:\n", i + 1);
        printf("Is account active? (1-Yes, 0-No): ");
        scanf("%d", &isActive);


        if (isActive == 0) {
            printf("Account inactive. Skipping...\n");
            continue;
        }

        printf("Enter plan (Basic/Standard/Enterprise): ");
        scanf("%s", plan);

        printf("Enter used storage (in GB): ");
        scanf("%f", &usedGB);

        float planLimit;


        if (strcmp(plan, "Basic") == 0) {
            planLimit = 100.0;
        } else if (strcmp(plan, "Standard") == 0) {
            planLimit = 500.0;
        } else if (strcmp(plan, "Enterprise") == 0) {
            planLimit = 2000.0; // 2 TB = 2000 GB
        } else {
            printf("Invalid plan entered.\n");
            continue;
        }

        float remaining = planLimit - usedGB;
        float extraCharge = 0.0;
        float usagePercent = (usedGB / planLimit) * 100;

        if (usedGB > planLimit) {
            float extraGB = usedGB - planLimit;
            extraCharge = extraGB * 4;
            remaining = 0;
        }


        printf("Plan: %s\n", plan);
        printf("Used Storage: %.2f GB\n", usedGB);
        printf("Remaining Storage: %.2f GB\n", remaining);
        printf("Extra Charges: Rs. %.2f\n", extraCharge);


        if (usagePercent > 100) {
            printf("Status: Storage LIMIT EXCEEDED! Extra charges appl
        } else if (usagePercent >= 95) {
            printf("Status: WARNING! Storage usage exceeds 95%%.\n");
        } else {
            printf("Status: Normal usage.\n");
        }


        serverUsed += usedGB;


        if (serverUsed >= serverCapacity) {
            printf("\nServer storage is FULL. Stopping further process
            break;
        }
    }

    printf("\nTotal Server Storage Used: %.2f GB out of %.2f GB\n", se

    return 0;
}
```

Run | Save

Input

```
3
1
Basic
95
```

Output